

Lesson 20 — Design Challenge — Smart Guard

Matches LEGO Teacher Portal lesson: "Ride Rush".

Objective (I can...): I can design a model that makes its own decisions using conditionals. **CS concept:** Applying if/then/else with sensors · **Builds on:** Lessons 16-19. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): A smart doorbell decides: IF it sees a person, THEN alert; ELSE ignore. *First-principles question:* "What sensor + rule would make YOUR guard smart?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, trigger objects. **Model already built** (Conditionals model from Lesson 16) — the 30 min is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Conditionals review — pairs match scenarios to if/then vs. if/then/else. *(Theme: Maze/guard face-off.)* - **Concept + demo (10):** Challenge: build a "smart guard" or "helper" that decides between two actions using a sensor. Coach models a pseudocode plan with a condition, then hands off. *(Rotate the format: try "coach makes a mistake, kids catch it.")* - **Build/Code (30):** Pairs plan and code an if/then/else decision driven by a sensor on the built model. *Swap at 15 min.* - **Challenge (7):** Add a second condition or combine with a loop so it decides repeatedly. - **Share + pack-up (5):** **Two ways to solve it** — pair up two teams that met the *same* goal with different working programs. Each shows theirs, then together they decide which solution is simpler and say why (there's often more than one right answer). **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks the condition; swap at 15 min. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a one-condition target with a partial plan. *Extension / fast-finisher* — require a loop plus if/then/else. *Reteach* — reuse a Lesson 18-19 program as a template.

Exit check: Pair delivers a model that chooses between actions correctly using a condition. **Progress layer:** Send home the **Unit 4 parent note** ("your child can make a model decide on its own") with the **build photo**. **At-home (optional):** Invent a "smart" gadget; write its main if/then/else rule. ~10 min.
